

Remote Monitoring and Medical Devices Control in eHealth

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Abstract— This paper presents a solution that improves remote monitoring and real-time control of medical devices in eHealth, and provides better user experience and deployment simplicity. Our approach builds a unified, scalable, secure and real-time platform making use of IoT Gateways to remotely monitor patients and manage/control diverse types of sensors and IoT medical devices. WebRTC is used as a key architectural component to enable interactive communication between smart medical spaces (e.g., clinics, patients' homes) and remote users (e.g., health/wellness services, physicians, medical staff). We also demonstrate the need for an Edge Cloud to save on network bandwidth usage especially for video traffic for remote monitoring and to ensure real-time control of medical devices. We implemented a Proof of Concept (PoC) to test our solution. The performance results obtained measure the utility of the proposed approach and its effectiveness in real-time eHealth services.

Keywords— eHealth; IoT; Big Data, Smart Medical Spaces; Real-time Services, Remote Monitoring; WebRTC, Medical Devices Remote Control; Edge Cloud

I. INTRODUCTION

The Internet of Things (IoT) has tremendous potential for disrupting the current Internet architecture. Conservative estimates predict that by 2020, over 50 billion devices will be connected to the Internet [1]. Additionally, 40% of all of the worlds' data purportedly will come from IoT devices and sensors [2], which are expected to generate a total of 129 yottabytes [3]. To put this much data into perspective, consider that 90% of the data in the world has been created in the last two years and the creation of data will continue to accelerate [4]. As a result, functionality that would normally reside in a backend cloud is migrating to be more proximate to IoT devices and to the location where data is created. In fact, the IDC predicts that within two years, over 40% of all IoT created data will be stored, processed, analyzed, and acted upon at the edge of the network [5]. This has given rise to Edge computing, to ensure that the right processing, storage and networking occurs at the right place and right time. It has been shown that the cost of using local resources in conjunction with remote cloud resources can result in a solution that is one third the cost of the remote cloud-only cost [6]. Therefore Edge and Cloud computing will need to co-exist.

With the promise that billions of devices will connect to the Internet and will lead to an immense number of different

IoT applications, several advanced services are emerging and among them eHealth is representing a revolutionary segment. Body Area Networks (BAN) is one trend that has been discussed for some time and makes use of different sensors attached to the body and connected together to provide useful health information or to trigger an action related to medication. Remote monitoring of patients is another trend that is increasingly popular and has already witnessed attention from real Internet service providers. Robotic surgery is another revolutionary trend that allows remote surgeons to perform minimally invasive surgeries. All this requires real-time video monitoring and real-time control of medical devices and systems.

However there is not yet a unified solution for easy setup, remote control, real-time interaction, nor a scalable solution for management of large numbers of sensors and medical devices, not to mention their data. Current IoT monitoring/surveillance and remote control solutions have several limitations, which are mainly as follows: (1) Each sensor system within each IoT solution has its own software application and its own communication infrastructure to the back-end the Cloud (for sensor readings and sometimes control, such as with the Nest thermostat, Tesla EV, Fitbit wearable). This results in a tedious process for the user who must install many different applications, as well as inefficient network utilization for Cloud communication; (2) Almost all of those IoT devices run within the network firewall. Yet, there is no standard and manageable approach to connect and to communicate (2-way) with the external Cloud server or remote users who are monitoring the system, nor to share data across applications; (3) IoT solutions do not always provide real-time interactive communication to remotely monitor and control IoT devices and sensors (e.g., adjusting the device parameters, turning on/off the device, switching cameras on/off, panning, tilting and zooming, etc.); (4) Video surveillance solutions in the IoT are still limited because: (i) they do not always support low cost IP cameras and do not guarantee low-latency interaction, and (ii) they lack efficient video storage infrastructure for proactive real-time video analytics (not only communicating the video upon application request, but also sending asynchronous video alerts as needed for "illustrative notification" that include video evidence with the alerts).

We provide a solution that resolves the aforementioned limitations and that builds a unified, scalable, secure and

real-time platform. Our solution uses IoT Gateways to manage diverse types of sensors and medical devices, and it employs WebRTC to enable real-time, interactive and secure real-time communication between smart medical spaces and remote users (e.g., health/wellness services, physicians, medical staff). The rest of this paper is organized as follows: Section II discusses the related work and the motivation for our solution. Section III presents the details of our solution for remote monitoring in eHealth and interactive control of medical devices through a unified platform. Section IV, describes the Proof of Concept (PoC) that we implemented to demonstrate our design and Section V presents the performance evaluation as well as a comparison against popular solutions in the market. We conclude the paper in Section VI and discuss our future work.

II. BACKGROUND AND MOTIVATION

A. Related Work

Although monitoring and surveillance are often coupled together as if they are one unified topic, they address two different but related use cases. Surveillance describes the act of observation using video and typically is associated with security. Monitoring on the other hand is the generic term for collecting data (video or non-video) and is not necessarily associated with security. Few literature contributions deal with monitoring and surveillance in one solution. The monitoring and surveillance solutions mainly address three sectors: smart homes, eHealth and roads and public areas.

The work in [7] explores the different implementations of IoT in home automation for energy and water management, and comfort and conveniences. Different technologies (like IPv6 and Cloud computing) are reviewed. The notion of a centralized intelligent hub for home control is presented and the approach of using standardized protocols is highly recommended. The authors in [8] propose an architecture to enable the users to control and monitor smart devices remotely based on GSM wireless communication from a web server into the smart home. In this architecture, the user sends commands through the web which are converted into GSM-SMS commands and sent to a GSM embedded module in an IoT agent in the smart home. Although, this solution allows remote control, it is limited to a GSM architecture and it does not support video monitoring or 2-way communication.

The work in [9] develops wellness sensor networks for smart home systems. This tracks real-time data related to the object usage and movement inside the home, to forecast the wellness of an individual. Only Zigbee sensors are used in this solution, transmitting their data to a Zigbee dongle attached to a PC. This limits the solution scalability and requires the constant presence of a PC. In addition, this solution lacks video monitoring and computer vision that can add value to wellness tracking. The authors in [10] highlight the opportunities and challenges for IoT in the future of health care. Networked sensors, either worn on the body or embedded in the living environments, make possible the gathering of rich information indicative of physical and mental health. However, it is not clear where this gathered data is processed and analyzed or what action is triggered

upon data anomaly. On the other hand, [11] builds an Android platform for the healthcare domain with an application that provides Electro Cardiogram (ECG) waves visualization to the end user together with data logging functionality in the background. The logged data can be uploaded to the user's private centralized cloud or a specific medical cloud, which keeps a record of all the monitored data and can be retrieved for analysis by medical personnel. This solution necessitates the presence of the Smartphone as a companion device and is OS-limited. Also, the constant transmission of sensitive personal data to the Cloud may not be easily accepted by all users.

In terms of surveillance, [12] presents a surveillance system for the management of road vehicles through the extraction of the vehicle visual tags (that can be license plate number, vehicle color, and vehicle type) and applying computer vision algorithms to recognize vehicles. In this solution, the computer vision operation takes place in the Cloud, which requires continuous transmission of video frames or images for the visual tags to the Cloud that makes the solution limited to regions and geographical locations that support bandwidth-rich network infrastructure along roads. The work in [13] considers surveillance of International border areas, which can be hostile environments with mounted fixed view cameras covering very limited areas. A solution is presented using a wireless remotely-controlled robot with a mounted camera to detect the images of living human figures with the help of Passive Infrared (PIR) sensor for rescue operations. The information regarding the detection of live humans is given on a webpage from the PIR sensor. In addition, the user is able to access the video transmission from the robot and the camera mounted on the robot can be controlled remotely to move horizontally and vertically. Despite the interest of this solution in rescue operations, it is not clear how the robot will deal with the continuous video captured by its mounted camera. Continuous transmission of this video to the Cloud is not a practical approach especially for a hostile environment. In addition, the wireless communication reliability cannot be ensured, whether or not in a hostile environment.

From our investigation of the literature on monitoring and surveillance, we noticed the lack of a unified platform for both monitoring and surveillance and that provides interoperability with different IoT sensors, devices and IP cameras. Also, most solutions depend on near constant communication with the Cloud and lack support for 2-way real-time communication. Our solution aims to resolve these limitations and to provide a reliable remote monitoring and control solution fitting eHealth applications and real-time services. IoT Gateways in our solution provide interoperability with a plethora of IoT sensors, medical devices and IP cameras and allow data and video aggregation and processing at the edge. We also use WebRTC (Web Real-time Communication) [14] for interactive 2-way real-time communication. As the analysis forecasts, over 6 billion WebRTC-enabled devices and over 2 billion WebRTC active users are expected by 2019, where WebRTC is quickly ramping up to become a ubiquitous mainstream platform for

real-time communication, not only for Web communication, but also for mobile and IoT device communication [15].

B. Motivation and Approach

Current solutions for IoT smart spaces lack a unified platform to control and to manage the many IoT sensors and devices, their heterogeneity and how video applications are handled. Without a unified software platform, it is difficult to impossible to support 2-way communication from devices to remote applications and vice-versa. This impacts the mass and scalable deployment, as well as cost efficiency of eHealth applications. Additionally, video monitoring in IoT that uses IP cameras have had very few solutions until recently and most require the IP cameras to connect directly to the Internet to stream video to the Cloud for processing and analytics. This approach consumes a large amount of network resources to continuously transfer video over the Internet and requires eHealth application to open the smart medical space to the Internet to expose the IP cameras to the external world, which can present security risks or require separate security solutions.

In contrast, our solution was designed to ensure interoperability as well as reliable and secure communication between users (e.g., patients, elderly) in smart medical spaces and remote users (e.g., health/wellness services, physicians, medical staff) through:

- Interoperability with different types of IoT sensors, medical devices and IP cameras using an IoT Gateway that not only plays the role of an aggregator to data and traffic, but also provides a service abstraction to support the different sensors/devices communication protocols;
- Real-time interactive communication to remote users through WebRTC;
- Advanced processing and storage capabilities at the Edge of the network (near the data sources) to deal with the real-time data processing, analytics and storage, especially video that has the potential to generate heavy workloads and patients information that can be sensitive to share globally;
- Ability to scale up to connect multiple sensors/medical devices of different types (reading different types of data) through the same IoT Gateway and thus providing a cost-effective solution for mass deployment;
- Native security mechanisms through WebRTC (that offer secure and encrypted media channels) without opening the whole home network to the Internet;
- Privacy preservation since the sensitive data are analyzed locally and only notifications are sent to remote users. Also videos are stored locally

on the IoT Gateway and only sent to the remote user for monitoring on-demand and through a secure media channel provided by WebRTC;

- Ease of access and cross-platform user experience through the availability of web interfaces.

III. REMOTE MONITORING AND MEDICAL DEVICES CONTROL

Our unified platform solution uses a central communication hub to provide an abstraction layer to control and to aggregate data and video from multiple diverse sensors, medical devices and IP cameras. Figure 1 gives an overview of the end-to-end solution and the following sub-sections detail the solution enabled by the technologies.

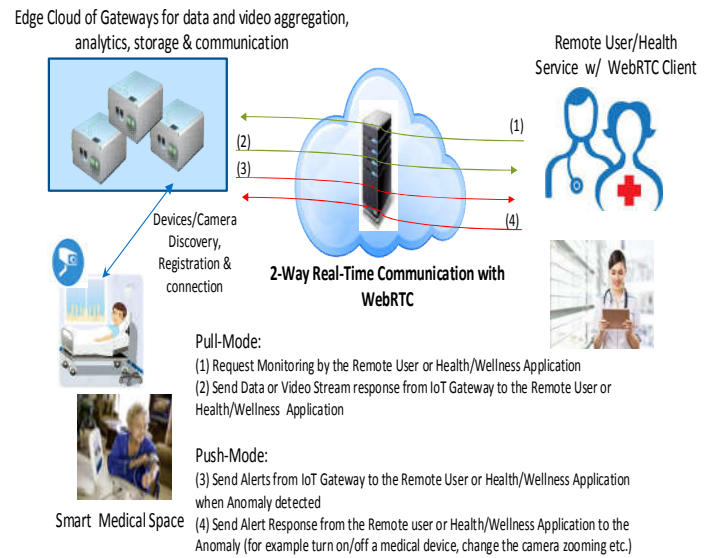


Figure 1. Overview of the End-to-End Solution

A. Edge Cloud Capabilities

An Edge Cloud provides efficient communication and storage infrastructure for body sensors, medical devices, and IP cameras and serves as a proxy to the back-end Cloud, if and when needed. The IoT Gateway plays a critical role in the Edge Cloud as it serves not only as a communication hub, but also as an aggregation hub for local Video/Data storage, processing and analytics. Because the Edge Cloud is more proximate to the Video/Data sources, it offers: (1) A smaller ecological footprint especially for video workloads that generate heavy traffic; (2) Push-mode remote monitoring (e.g., if an anomaly is detected, illustrative sensor data or a brief video clip can be sent with the alert); (3) Pull-mode remote monitoring support with smart data and video retrieval (e.g., remote users can request current/past data or videos for a specific event and named descriptions); (4) Improved privacy enabling for stored data and videos.

B. WebRTC (Web Real-Time Communication)

For real-time video session management and remote control of body sensors, medical devices and IP cameras, WebRTC (Web Real Time Communication) is used and its standardized APIs and protocol stack are introduced in the IoT Gateway. WebRTC enables 2-way secure and real-time communication (with Firewall traversal and encrypted sessions), a data channel that is useful for remote control commands, low-cost video codecs (VP8, VP9, and OpenH264 that are royalty-free), and first-class support for one-to-one and one-to-many communication.

As WebRTC is based on a client-server communication model, the IoT Gateway plays the role of a WebRTC client (integrating the WebRTC client SW stack) to aggregate data from multiple sensors, medical devices and IP cameras and to communicate with the WebRTC server hosted in the Cloud. The remote user device (phone, laptop) or 3rd Party Service uses WebRTC through a client application or a web browser (Chrome, Firefox) to connect to the IoT Gateway through establishing a WebRTC session managed by the WebRTC Server.

C. End-to-End Architecture

Our solution offers a unified end-to-end SW platform for remote monitoring in eHealth and real-time control of medical devices, leveraging Intel's HW. In this unified platform, the sensors, medical devices and cameras send their data or captured video to the IoT Gateway through standardized protocols using a compressed IP stack and efficient signaling: (i) CoAP for sensors management and low-latency data transmission and (ii) RTSP (Real Time Streaming Protocol) over SRTP (Secure RTP) for video in push or unsolicited mode (i.e., instantaneously) and also in pull or solicited mode (i.e., following a trigger from the remote user or health/wellness service through the IoT Gateway).

The IoT Gateway is running a WebRTC SW stack and starts the WebRTC session with the remote user or health/wellness service for monitoring, control and/or management. Multiple medical devices equipped with multiple sensors can be connected to the same IoT Gateway. Any remote user or health/wellness service, with authorization, can remotely access and control those appliances through a web interface provided by WebRTC. Figures 2 to 4 present the deployment architecture in different modalities. In this architecture, the IoT Gateway acts as a communication hub, data and video aggregator and is the embodiment of the smart Edge Cloud.

1) Remote Connection to Cameras & Medical Appliances: The remote user or health/wellness service connects to smart appliances in smart medical spaces (that can be a sensor, medical device/ or IP camera) to request Data or Video Streams on-demand. Figure 2 illustrates how Data or Video is continuously streamed to the IoT Gateway (Step 1 in Figure 2) and are locally stored and analyzed through the analytics algorithms sitting on the IoT Gateway (since the real-time video analytics cannot wait for roundtrip with storage servers residing at the Cloud). Storage can be

also on another local machine connected to the Gateway (e.g., server for long duration storage). When a remote user or health/wellness service requests a Data or Video stream, a WebRTC session is initiated with the IoT Gateway (Steps 2-3 in Figure 2) to serve up the Data or Video.

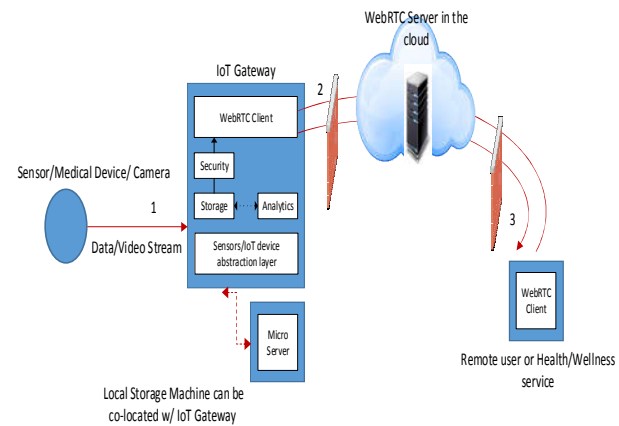


Figure 2. Remote User Request for Data and Video for Monitoring

2) Alert Notification to Remote Users: The IoT Gateway runs analytics algorithms to inspect the received Data or Video streams (step 1 in Figure 3). Upon the detection of any data anomalies or abnormal behavior in the video or the received sensors' data, an alert is sent to the remote user through SMS, email, Chat App (step 2.a in Figure 3) or active WebRTC session sharing the same Data or Video subject of the alert (step 2.b in Figure 3). The remote user can join the designated WebRTC session, retrieve stored or real-time video and check the status of the user or medical device directly.

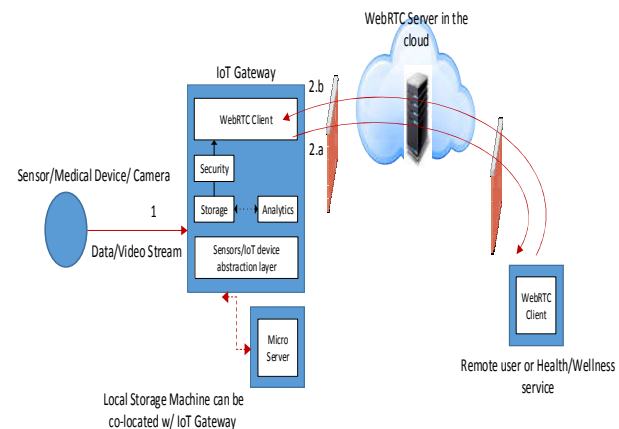


Figure 3. Alerts Notification upon abnormal conditions

3) *Remote Control of Medical Appliances*: A remote user can control smart appliances (e.g., turning on/off the camera, changing the camera angle, camera zooming, turning on/off a respiratory device, etc.) through using a WebRTC session with the IoT Gateway to transmit the control command via the WebRTC data channel (step 1 in Figure 4). The IoT Gateway then controls the camera or medical appliance (step 2 in Figure 4) through a direct communication.

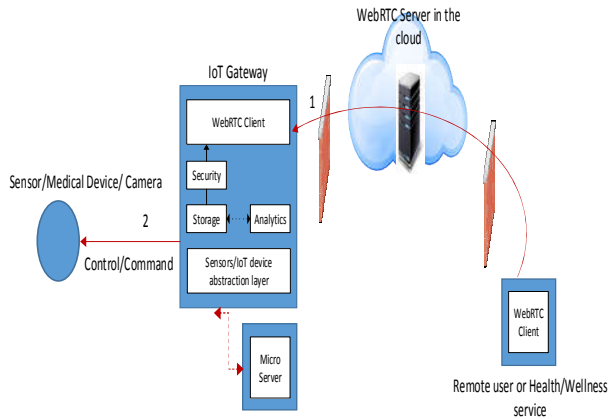


Figure 4. Remote Control of Smart Appliances

4) *Data Aggregation, Processing and Storage*: As illustrated in Figure 1, the solution uses the Edge cloud for storage and real-time processing for Data and Video streams. The IoT Gateway in this context has the role of data and video aggregation and storage management. Storage can be completely on the Gateway or on another local machine (server or micro-server collocated with the Gateway). In addition, the IoT Gateway performs Data and Video streams processing and analytics.

A. In the case of push mode from the sensors/medical devices and cameras, the Gateway receiving the Data or Video stream first runs analytics to extract trends and sends alert notifications on demand (i.e., when there are anomalies or “a certain behavior” noticed in the Data or Video stream, which can be decided according to pre-defined or configurable rules in the analytics algorithm or inferred over time when “normal” behavior is established). The notification function is performed by sending the Data/Video streams themselves showing irregular behavior, a simple message explaining the status or sending both.

B. In the case of pull mode from the sensors/medical devices and cameras, the Gateway receiving the Data or Video stream sends the stream directly to the remote user or Health/Wellness service through the WebRTC server (e.g., Video right now showing what is going on

in the patient room at the hospital, current Temperature/Pressure of the patient, if the respiratory device is On or Off, the status of heart beats, etc.). This Pull mode takes place on demand triggered by a request from the remote user or an alert notification received by the remote user.

In both push and pull modes, storage continuously takes place in data batches or video chunks or for pre-defined intervals. In addition, persistent storage takes place for data and video showing irregular events for later reference that might be needed. The IoT Gateways can generally support from 100G to 500G storage space on average. This means video storage capacity of 50G to 250G for 50% storage occupancy, implying many hours of HD videos.

IV. PROOF OF CONCEPT (PoC) DESCRIPTION

We implemented a Proof of Concept (PoC) for the end-to-end solution. Figure 5 illustrates the different software components deployed on the IoT Gateway.

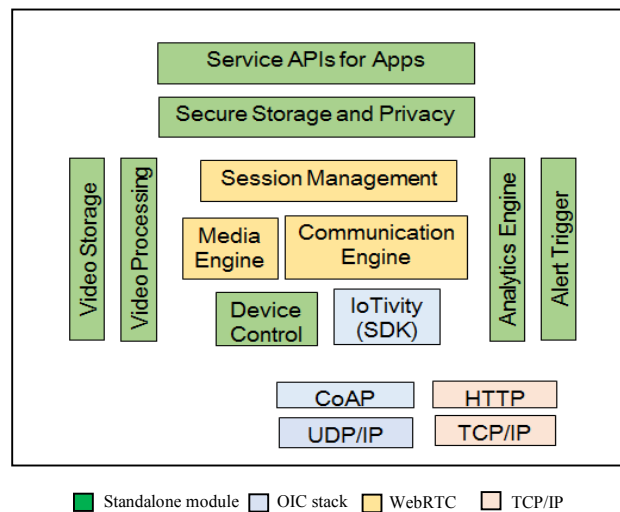


Figure 5. SW Architecture Modules in the IoT Gateways

We implemented the following modules in the IoT Gateway:

- **IoTivity [16] SW stack**: allows the automatic discovery and registration for OIC [17] devices (sensors and medical devices). We are also using the plug-in mode with IoTivity that allows interoperability with different types of devices (i.e., making non-OIC devices “including sensors and IP cameras” OIC-compliant).
- **Device Control**: we implemented this module to control multiple devices simultaneously. It intercepts with the IoTivity SW stack for device discovery, it stores device status in files, easing device access, and it provides an interface to all SW modules to connect to any device and also to the remote client (or any 3rd party IoT

application) to connect to the devices and control them.

- **WebRTC SDK:** we are using the WebRTC SW stack developed by Intel [18] and optimized for Intel Architecture (IA). WebRTC client SW stack is implemented in the IoT Gateway and the WebRTC server SW is implemented on a server that is hosted in a separate network. This server can also reside in the Cloud of the service provider or can be hosted in the same smart space on the same local area network, or in the same IoT Gateway of the WebRTC client (depending on localization requirements).
- **Video Processing:** we implemented live transcoding making use of ffmpeg open source libraries [19] to allow video capture in different formats and frame rates. Our video processing modules also allow image extraction from video to serve image-based computer vision algorithms in the analytics engine.
- **Video Storage:** we implemented a “video chunk and store” algorithm that chunks on the fly the video streams based on defined time intervals and that stores video chunks in files (with each file storing a short chunk of video).
- **Secure Storage and Privacy:** the implementation of this module is still work in progress. However, video streaming security is ensured through the WebRTC encrypted media channel.
- **Service APIs:** WebRTC APIs are used for video feeds from a source (cameras for live Video and storage for recorded video) and for video streaming to remote client or clients. Restful APIs are used for communication between the different SW modules.
- **Analytics Engine:** This module is a general purpose analytics engine for integration with any analytics algorithms and is why we consider the solution a blueprint for re-use. For the PoC, we integrated and tested gauge analytics for checking the status of a medical device in a hospital and face detection and recognition analytics for patients’ identification.
- **Alert Trigger:** This module receives inputs from the analytics engine and sends alerts to the remote user upon detection of abnormal events. Besides the implemented modules in the IoT Gateway, the WebRTC server SW is implemented in an external server. As WebRTC client SW exists in several browsers, the remote user in our tests used his PC and Smartphone with any browser (e.g., Chrome or Firefox) to connect with the IoT Gateway, to receive video or to remotely control the devices.

V. EVALUATION OF OUR SOLUTION

1) Performance Evaluation

We evaluated the performance of our proposed solution in different usage scenarios and with different gateway

platforms (NUC box Broadwell *CPU - Corei5 - 2 Cores and 109G HD*, Desktop (DT) device Haswell *CPU - Corei7 – 4 cores and 500G HD*). We considered the following metrics: CPU load%, video startup latency, latency for devices remote control, and storage occupancy for different video types. Table I, illustrates the performance for video streaming from live cameras for 1 and 5 minutes of remote monitoring. The CPU load in this case includes video capture and streaming. The video startup latency is almost 2sec for all cases (which confirms our observed video quality of experience). The CPU load is a little more when using the NUC (more than 50%) versus the DT platform (around 50%) and is slightly more than that when the monitoring duration increases from 1 to 5 minutes with both platforms. Table II displays the performance for video streaming of 1 minute stored video chunks for past events remote monitoring. As the remote client Chrome browser uses VP8 codec, transcoding is done at the gateway to go from H.264 to VP8, which adds to the delay and processing load. The video startup time for remote monitoring conforms to the expected video streaming quality of experience.

TABLE I. LIVE VIDEO STEAMING FOR REMOTE MONITORING

Time	Gateway Type	CPU load %	Video startup latency
5 min	NUC	58%	~2sec
	DT	50%	~2.86sec
1 min	NUC	53%	~2sec
	DT	50%	~2sec

TABLE II. STREAMING OF STORED VIDEOS FOR REMOTE MONITORING

Time	Gateway Type	CPU load %	Video startup latency
1 min	NUC	19%	~2sec
	DT	15-20%	~2.2sec

TABLE III. STORAGE OCCUPANCY

Video Description	NUC Storage Occupancy%	DT Storage Occupancy%
Bright light – static – no person (size 2.9M)	0.00266	0.00058
Low light – static – no person (size 1.4M)	0.00128	0.00028
Bright light – static – no person - more objects (size 3.0M)	0.00275	0.0006
Bright light - moving person (size 6.8M)	0.00623	0.00136
Low light - moving person (size 5M)	0.00458	0.001
Bright light - person - less motion (size 4.7M)	0.00431	0.00094

Table III illustrates the storage occupancy% on both platforms with different video characteristics. Video features impacts video size: brighter videos and videos with motion or more details (e.g., persons, objects) show more size.

Although the DT platform has better storage compared to the NUC platform, the storage occupancy for all types of videos is shown to be reasonable: ranging (0.00094 to 0.00458 occupancy%) for 1 minute video chunks storage. This implies that 1 hour of video recording will cost (0.05% to 0.2%) storage occupancy for the HD used in our platforms. Finally, we tested the latency for device remote control through remotely connecting to the IoT gateway and sending pan and tilt update command to the cameras. The obtained latency is 20msec using both Gateway platforms which conforms with the real-time latency specification.

2) Evaluation Versus Market Products

Figure 6 provides an end-to-end comparison for our solution versus Google NestCam. Our solution shows to be comparatively better in terms of features, scalability and deployment cost.

	Our Solution	Nest Cam
Network	Aggregated on local gateway Do not transmit video outside all the time (low bandwidth) P2P based transmission when needed	All devices connected to cloud (more bandwidth consumption) Transmit video (HD) all the time to cloud
Storage	Local storage, various storage profiles possible Cost of hard drive only	Storage on Cloud \$10/month, \$30/month
Analytics	Local analytical software on GW Personalized analytical algorithms Combined inputs (video, audio, and others)	Analytical on the cloud Concerns on privacy issues
Multiple Devices Support	Unified interface for all types of IOT devices Support various IOT protocols	Nest only Camera only
Local UI	Personalized UI supported within WLAN Localized information disclosure	Everything go through cloud
IP Cam	Any ONVIF Camera	Drop Cam only

Figure 6. End-to-End Comparison with Google NestCam

Figure 7 presents a comparison of our solution versus other popular video monitoring/surveillance solutions in the market based on solutions features and the deployment environment. Our solution covers multiple deployment environments and offers enhanced features.

	Smart Home	Corporate/Enterprise	VMS**	Cloud Storage/Control	Edge Storage/Control	No Monthly fees for storage	Web Browser Access	Video Clip Notification	Remote Monitoring & Devices Control	Full Platform (SW, HW)
Milestone		✓	✓	✓						
ONSSI/Ocularis		✓	✓	✓	✓					
Google Nestcam/Dropcam	✓								✓	
Samsung SmartThings	✓			✓				✓	✓	
Our Solution	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓



Support



Partial Support

**VMS: Video Management Software

Figure 7. Comparison with popular Solutions in the Market

VI. CONCLUSION & FUTURE WORK

With billions of sensors, IoT devices and IP cameras connecting to the Internet, and an immense number of different IoT applications coming online, eHealth applications and services are in continuous evolution. Consequently, new architectures are needed to manage and remotely connect to smart medical spaces. We provide a reusable blue print for remote monitoring and real-time control of smart medical spaces. Our solution offers a unified, scalable, secure and real-time platform that resolves several limitations with existing solutions. IoT Gateways are used to manage diverse types of sensors and medical devices, and WebRTC is employed to enable real-time, interactive and secure real-time communication between smart medical spaces and remote users or health and wellness services. In addition, we demonstrate the need for an Edge Cloud to save on network bandwidth usage especially for video traffic and to ensure real-time processing requirements. A PoC has been successfully conducted to test end-to-end functionality, performance, and the utility of the architecture for an eHealth use case. As future work, we plan to consider an encrypted storage approach for greater security and privacy and to add more edge devices to act as an Edge cloud cluster for heavy processing. We also plan to carry out a more detailed performance evaluation to provide guidance on the platforms types for different use cases and workloads.

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